Ergodic trajectories for different GP hyperparameters

Problem Settings

- Grid size of 25x10 km
- Robot model = single integrator
- Controller runs at 1/12.0 Hz
- Measurements assimilated every 5 minutes
- Each simulation total time = 1.0 hours

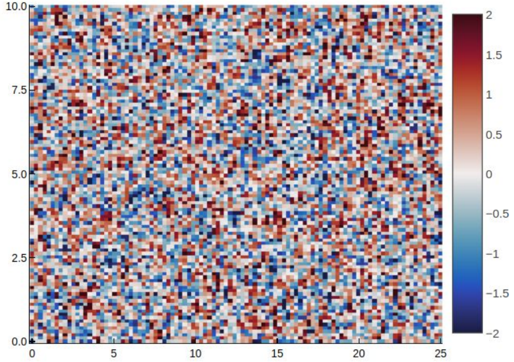
Spatiostatic (No time series)

Three Cases were evaluated

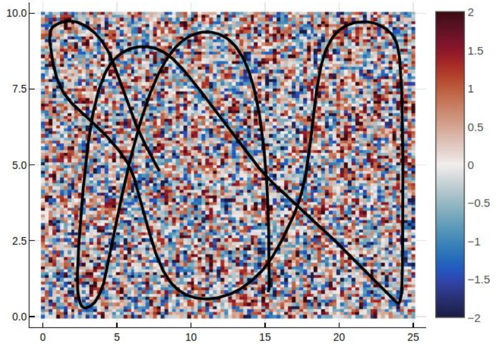
- Case 1
    - standard deviation (σ) = 1.0
    - spatial length scale = 0.1
- Case 2
    - standard deviation (σ) = 1.0
    - spatial length scale = 4.0
- Case 3
    - standard deviation (σ) = 1.0
    - Spatial length scale = 10.0

Case 1

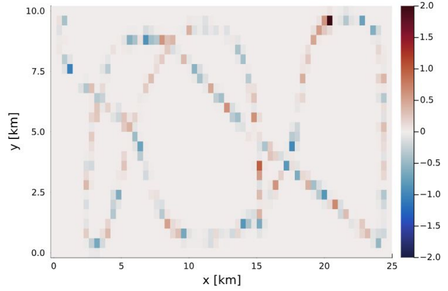
Domain



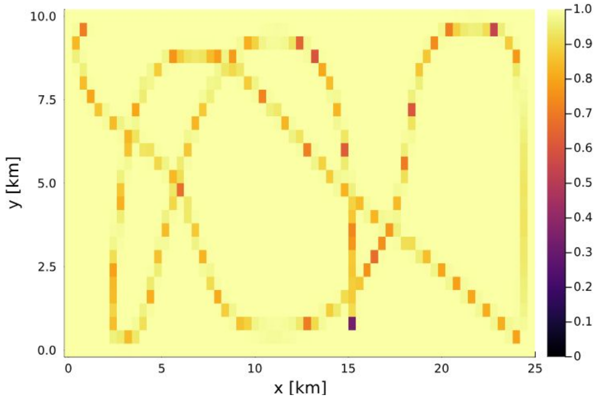
Trajectory



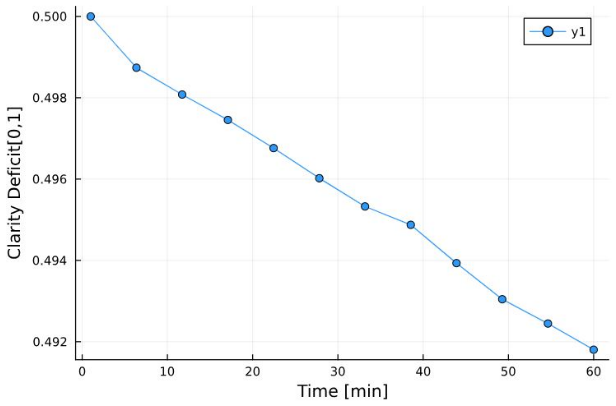
Environment estimation

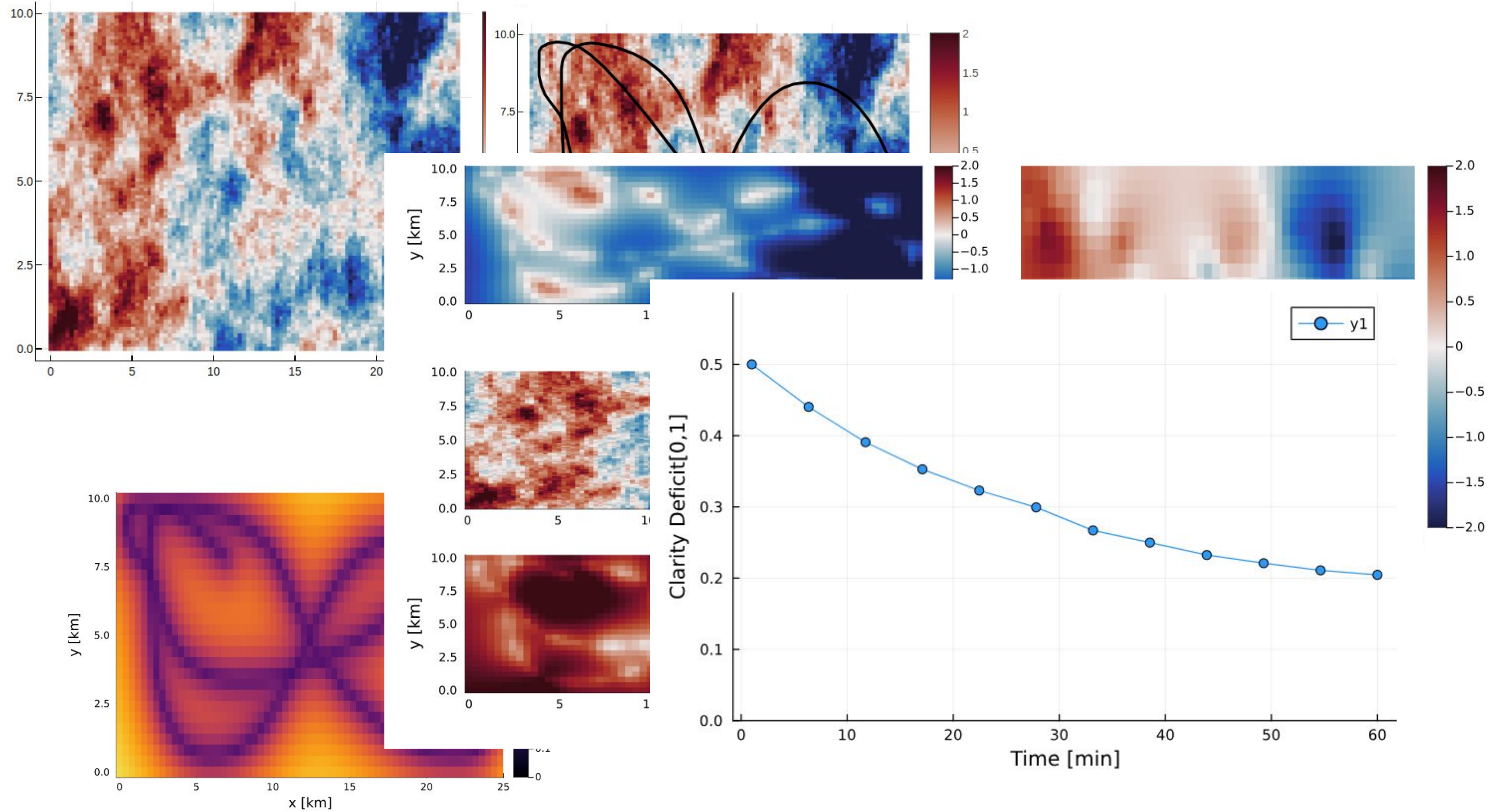


Kalman filter SD.



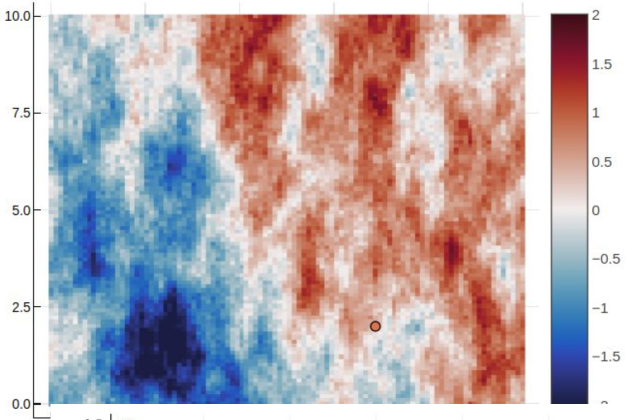
Clarity deficit

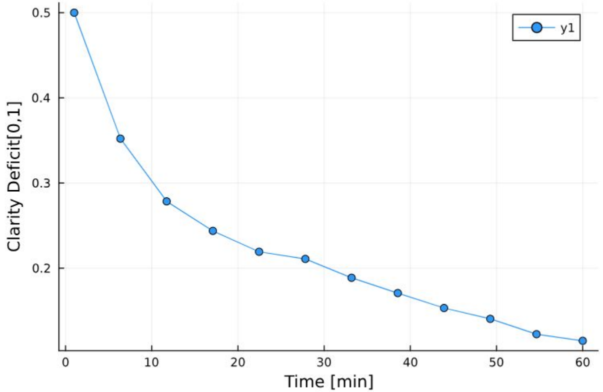


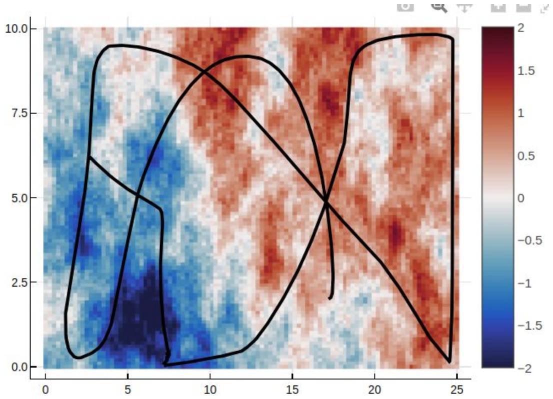


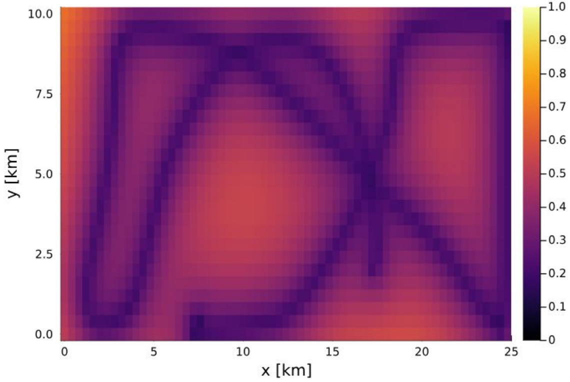
y [km]
x [km]
y1
Clarity Deficit[0,1]
Time [min]

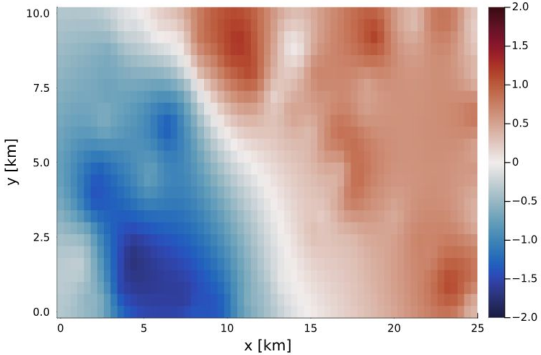
Case 3

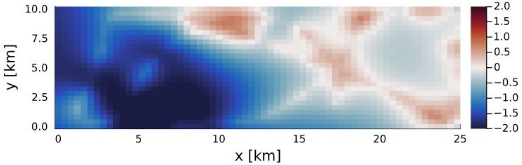


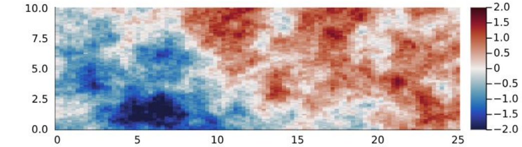


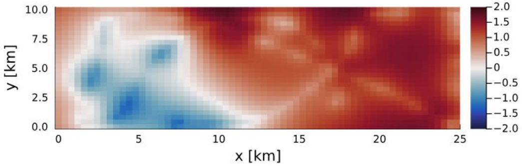










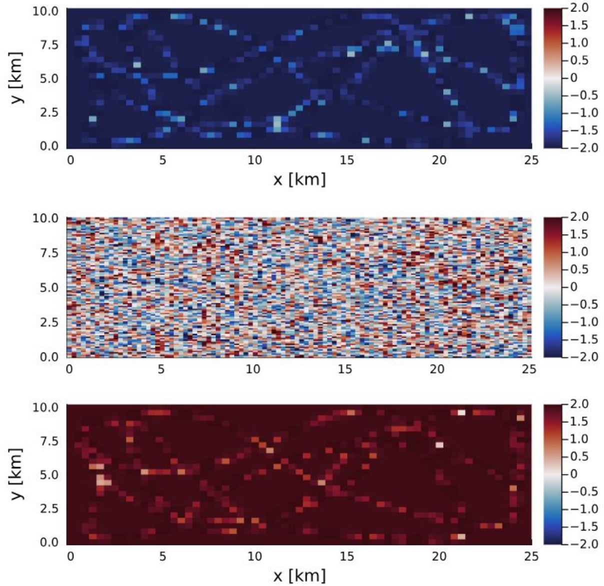


Spatiostatic (No time series) (Uniform TISD)

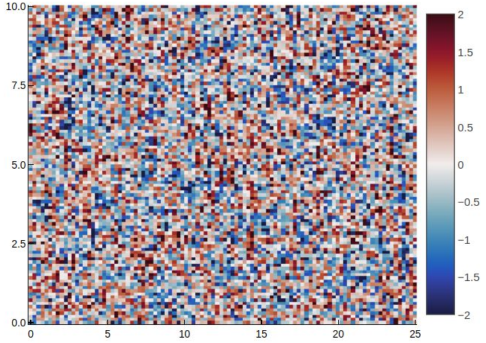
Three Cases were evaluated

- Case 1.1
  - standard deviation (σ) = 1.0
  - spatial length scale = 0.1
- Case 2.1
  - standard deviation (σ) = 1.0
  - spatial length scale = 4.0
- Case 3.1
  - standard deviation (σ) = 1.0
  - Spatial length scale = 10.0

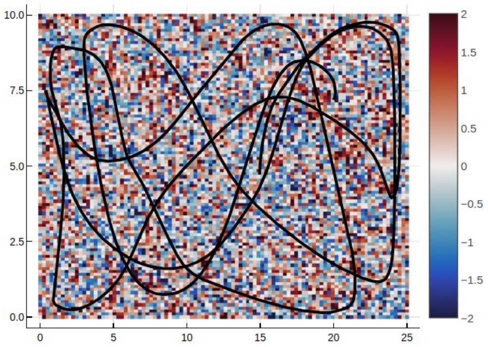
Case 1.1



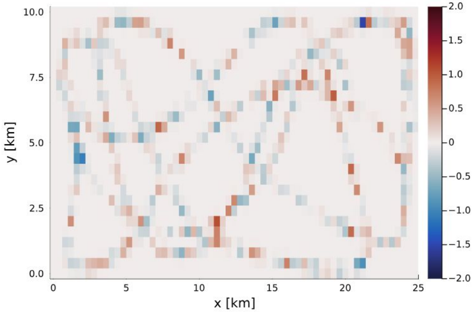
Domain 25x10 km



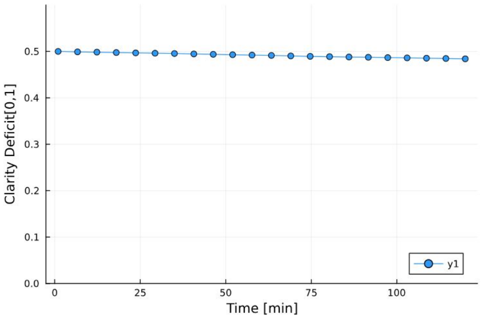
Single Integrator trajectory (2 hours)



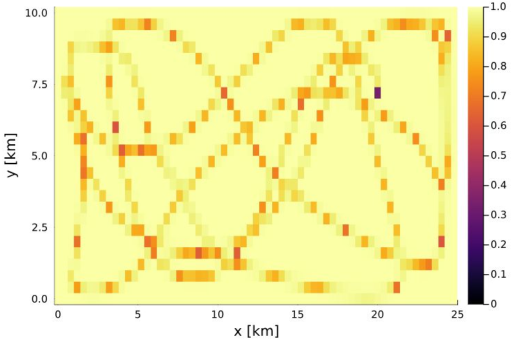
Environment estimation



Clarity Deficit

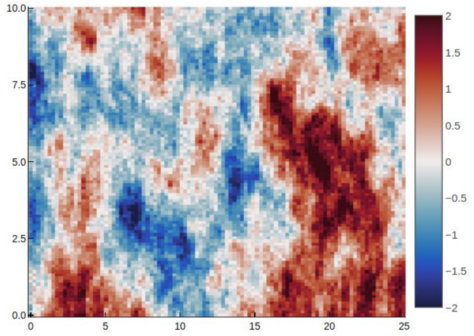


KF standard deviation

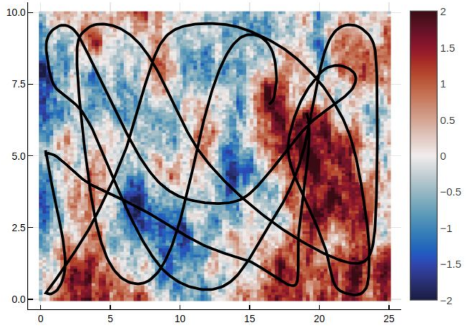


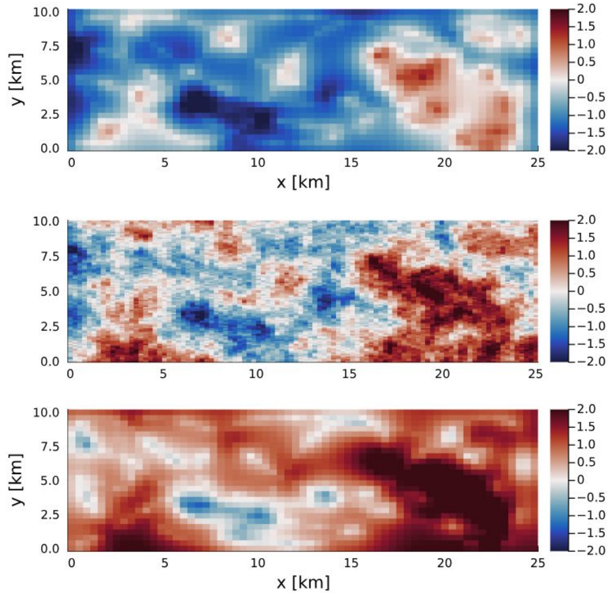
Case 2.1

Domain 25x10 km

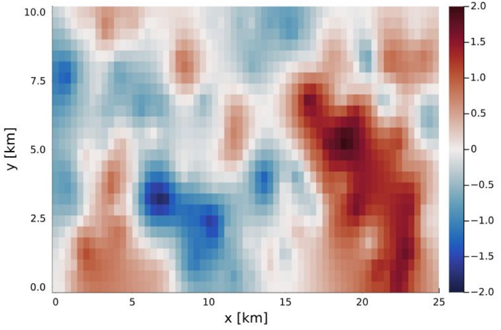


Single Integrator trajectory (2 hours)

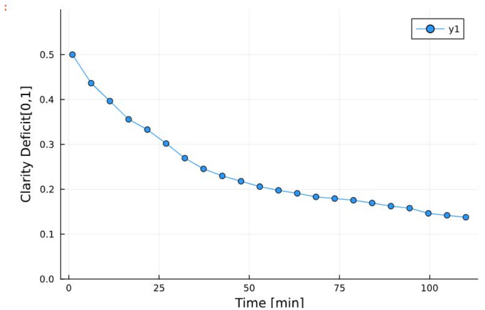




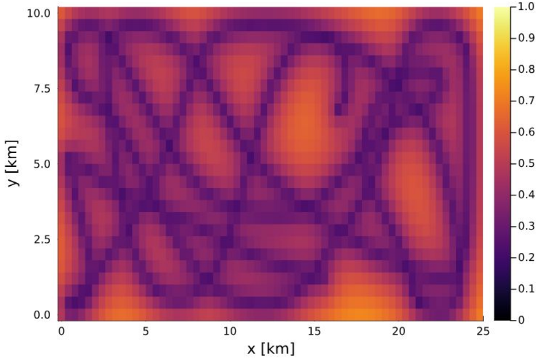
Environment estimation



Clarity Deficit



KF standard deviation



Case 3.1

Domain 25x10 km

Single Integrator trajectory
(2 hours)

Environment estimation

Clarity Deficit

KF standard deviation

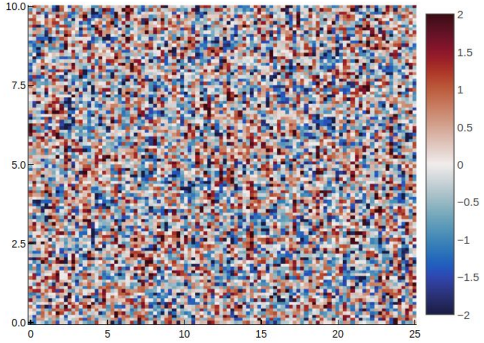
Spatiostatic (No time series)

Three Cases were evaluated

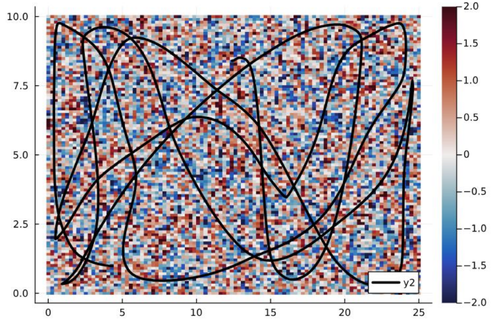
- Case 1.2
    - standard deviation (σ) = 1.0
    - spatial length scale = 0.1
- Case 2.2
    - standard deviation (σ) = 1.0
    - spatial length scale = 4.0
- Case 3.2
    - standard deviation (σ) = 1.0
    - Spatial length scale = 10.0

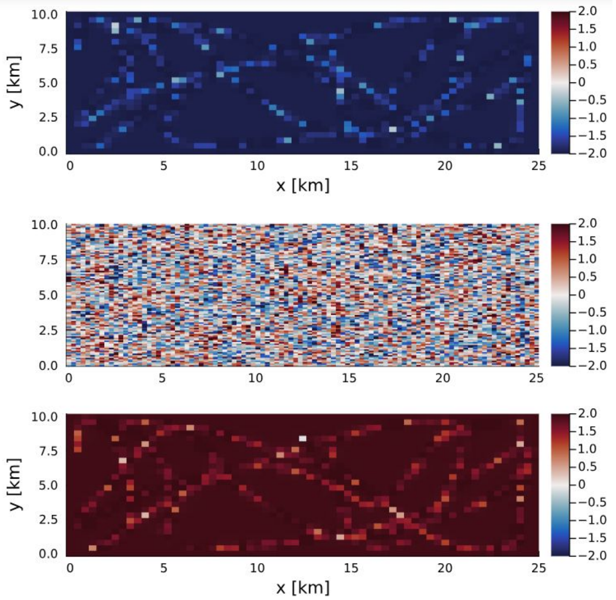
Case 1.2

Domain 25x10 km

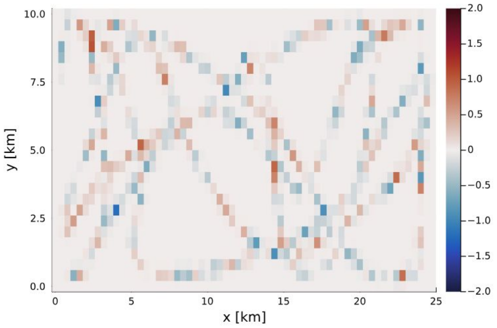


Single Integrator trajectory (2 hours)

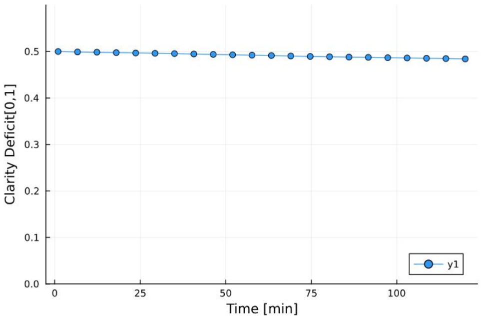




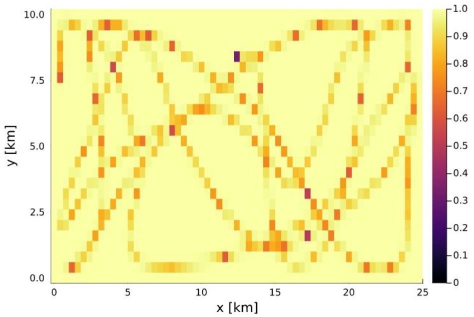
Environment estimation



Clarity Deficit

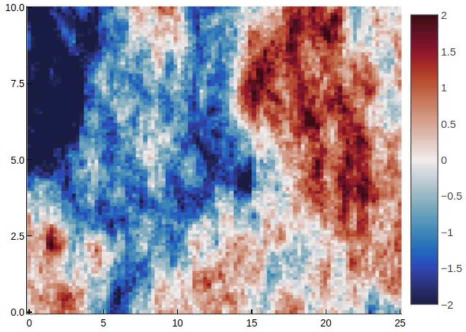


KF standard deviation

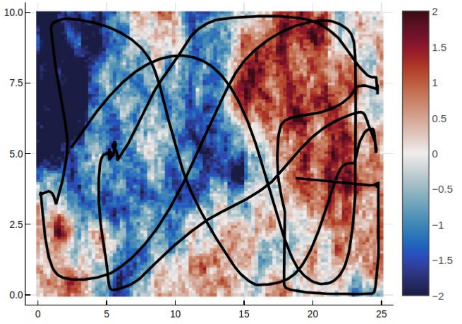


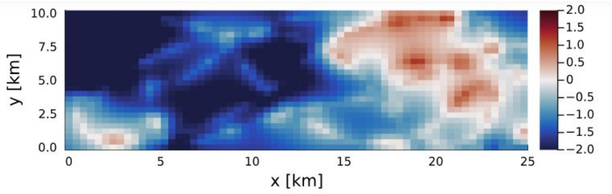
Case 2.2

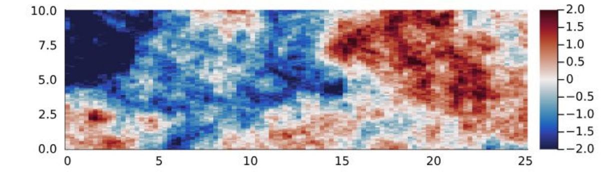
Domain 25x10 km

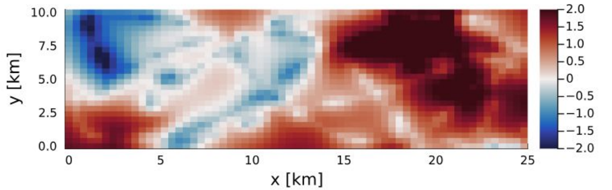


Single Integrator trajectory (2 hours)

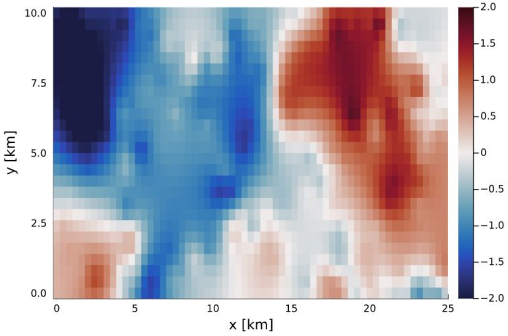




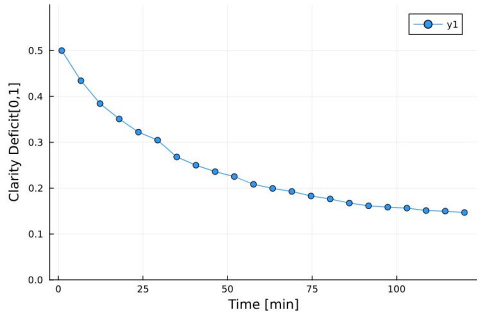




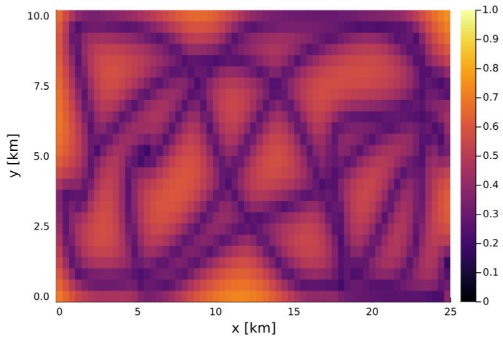
Environment estimation



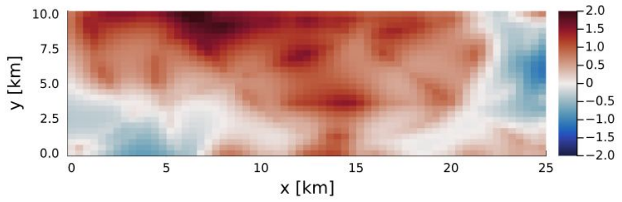
Clarity Deficit

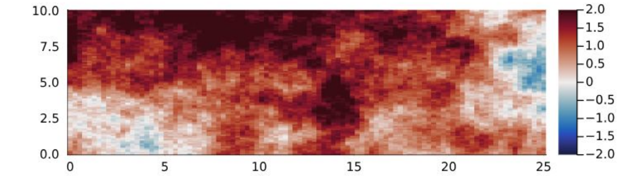


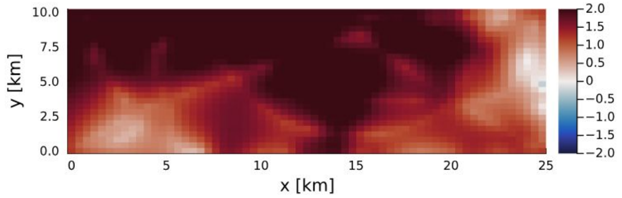
KF standard deviation



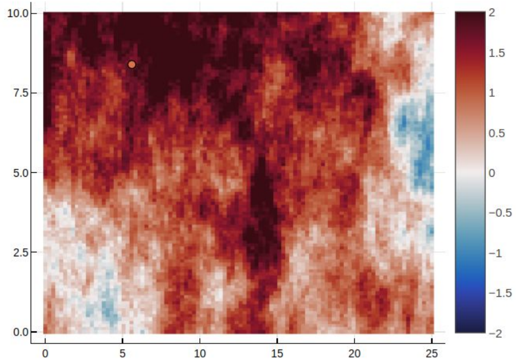
Case 3.2



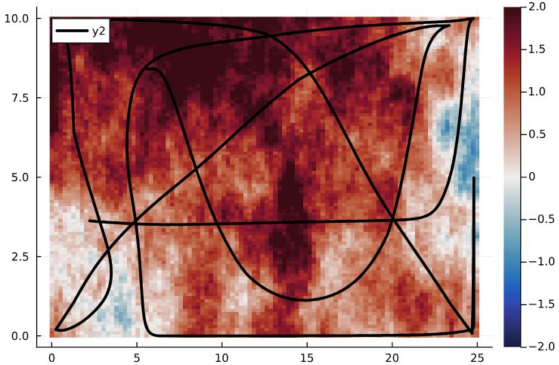




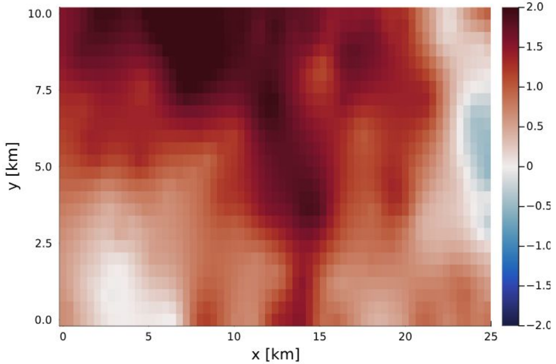
Domain 25x10 km



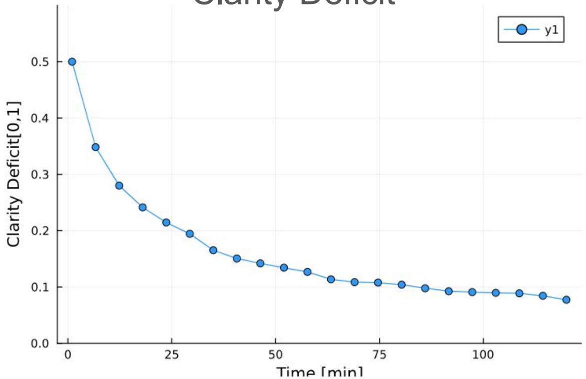
Single Integrator trajectory (2 hours)



Environment estimation



Clarity Deficit



KF standard deviation